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# Cognitive behavior therapy for academic burnout, procrastination, self-handicapping behavior, and test anxiety among adolescents: a randomized control trial

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## Abstract

**Objective** Academic stressors among adolescents, strongly associated with emotional disturbance, increase the chance of psychiatric problems, and their severity increases over time when emotional and educational issues remain unresolved. The present study is designed to investigate the impact of cognitive behavioural therapy (CBT) on procrastination, burnout, self-handicapping behaviour, test anxiety, and school functioning among adolescents facing academic problems.

**Materials and methods** A total of 200 adolescents were enrolled for eligibility assessment; 129 participants met the eligibility criteria, and they were allocated to the experimental and waitlist control conditions. After the baseline assessment, participants were involved in the treatment condition, and after 6–8 follow-up sessions, they were referred for the post-assessment. We used different assessment measures to assess the outcome, i.e. General Procrastination Scale (GPS), Maslach Burnout Inventory-Student Survey (MBI-SS), Self-handicapping Scale (SHS), Test Anxiety Inventory (TAI), and Living up to Parental Expectation Scale (LPES). Repeated measures ANOVA was used to analyze the results.

**Results** The current RCT findings suggest that CBT was found an effective treatment condition to address the emotional problems among adolescents. CBT significantly reduced the degree of procrastination {F (1, 63) = 25.01, $p < .000$, $\eta^2 = 0.29$} academic burnout {F (1, 63) = 11.08, $p < .000$, $\eta^2 = 0.16$}, test anxiety {F(1, 63) = 88.17, $p < .000$, $\eta^2 = 0.59$}, and self-handicapping {F (3, 56) = 10.17, $p < .000$, $\eta^2 = 0.16$} among adolescents. CBT also significantly helped the students to manage parental unrealistic expectations through providing relationship skills and training {F (3,56) = 546.46, $p < .001$, $\eta^2 = 0.89$). Further analysis reveals that counseling sessions substantially improved students' academic performance and students functioning at school in term of attendance and punctuality.

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**Conclusion** It is concluded that CBT efficiently addressed emotional and academic problems (i.e. procrastination, burnout, test anxiety, and self-handicapping behavior), improved students' functioning at school (i.e. attendance & and academic grades), and guided the students to manage unrealistic parental expectations.

**Keywords** Procrastination, Academic burnout, Test anxiety, Self-handicapping behavior, Attendance, Academic performance, Adolescents

## Introduction

Academic burnout strongly correlates with mental health problems and academic performance among adolescents. It also affects students' capacity to cope and manage academic tasks and promotes depressive symptoms among students [1]. Another important factor is academic procrastination, which increases the chance of burnout, emotional disturbance and educational failure. Initially, students use procrastination to get superficial relief, but consistency in behavior leads to academic problems [2]. In addition, academic burnout and procrastination promote tests and exams related to multiple anxieties and fears of losing grades [3] and academic outcomes [4]. Self-handicapping is another form of maladaptive behavior that increases the chance of burnout, procrastination and poor academic performance [5]. Furthermore, a significant positive relationship is investigated between test anxiety and self-handicapping behavior [6].

High self-handicapping behavior reduces the degree to which to put effort into the tasks. It enhances the expression of being more stressed before the exam and excessively worried about the exam's outcomes [7]. Self-handicapping behaviour among students promotes maladaptive behavioural patterns that superficially produce negative academic outcomes and, in later life, cause psychiatric problems [8]. Self-handicapping also correlates with procrastination [9]. Some studies explored there can be inverse relationship between procrastination and self-handicapping behaviour [10]. Self-handicapping behaviour is negatively associated with self-esteem, school achievement and teacher-student interaction and positively associated with adjustment problems, maladaptive behavioural patterns, and mental health problems [11].

Procrastination was also positively related to academic burnout [12, 13]. Some research has indicated that procrastination is related to depression [14], higher stress [15, 16], increased illness, and stress [17], higher anxiety [18], less effort on the task [19], boredom proneness [20], and poor academic performance [21–24]. In short, academic procrastination affects students' psychological well-being and academic achievement. Procrastination is a tendency to avoid academic tasks [25]. It negatively impacts a student's learning, attitudes, and academic performance, which usually changes into academic failure when intensity increases over time [26]. Moreover, academic burnout and procrastination positively associates with unhealthy behaviours [27]. For example, smoking, drinking, insomnia [28], anxiety [29, 30], depression [14, 30], poor self-confidence [30], and low quality of life [31]. High procrastination correlates with difficulty in managing academic goals [32].

According to an estimate, procrastination occurs in 46% of writing tasks, 30.1% in reading, 27.6% in studying for exams, 23% in attending classes, 10.6% in administrative task, and 10.2% in entire academic performance [33]. Burnout and procrastination make 53% of students vulnerable to psychological distress in a sample [34]. In Turkey, burnout cause depression (27.1%), anxiety (47.1%) and stress (27%), respectively [35]. Moreover, in Canada, students perceived psychological issues 30% [36], and in Malaysia, 41.9% [37]. In Indian students, 51.3% depression, 66.9% anxiety and 53% stress [38] and in Pakistan, 7.5% of students perceived a low level of stress, 71.67% moderate level of stress and 20.83% high level of stress [39] and medical students were found at high risk [40].

Parenting practices (e.g., warmth, harsh punishment) and parental involvement (e.g., helping with homework, communicating with teachers, volunteering at schools) reduce students' academic burnout and enhance academic involvement [41]. Moreover, healthy parental expectations promote relationship involvement, support, and encouragement, while irrational expectations are usually associated with emotional problems in students due to fear of not fulfilling the parental demands and expectations [42]. Rational parental demands, positive involvement and logical expectations are significantly associated with higher grades [43], academic motivation [44, 45], higher grades [46, 47], and better school attendance [48]. A positive parental attitude encourages students to work hard and put in emotionally stable and comfortable ways [49], while negative parental attitudes enhance mental stressors and academic failure [50].

After collecting some evidence, it is observed that the intervention-based programs remain successful in managing emotional and academic problems among adolescents [51]. For example, one of the intervention programs produced reasonable outcomes to address emotional disturbance among adolescents [52]. Similarly, CBT is found to be an evidence-based intervention to address procrastination using the conceptual framework, addressing negative thoughts and irrational beliefs [53]). In addition, CBT facilitated the students to cope with academic problems using different skill training and problem-solving methods to address academic issues [54, 55] and effective

student-teacher interaction [56]. The program includes techniques such as recording dysfunctional thoughts, directed exploration, Socratic questioning, relaxation exercises, homework and role-playing. CBT was more effective for self-handicapping behaviour and increased academic performance [57, 58]. Furthermore, interventions successfully support to management of self-handicapping, which is positively linked with procrastination and academic problems [59].

In Pakistan, merit competition for higher education is rising high every year and in the medical and engineering fields, this is more stressful. This has become an environment in academic institutions where only high scorer students can win the competitions. In this run, private institutions are producing better results through strict environments, prolonged study hours and other strategies to give better scores. Unemployment, per-pressure and parental unrealistic expectations are the other major sources that associates with academic stress and emotional disturbance. Most of the studies have been conducted in technically advance countries, where they have good infrastructure, facilities and enough resources to accommodate the students but in case of the lower/lower-middle income countries this situation is totally different. Similarly, in Pakistan, the cultural studies that explore mental health problems are available but an intervention-based studies are not available that will address the ground realities and give us help to aware and guide the students, parents and teachers, which is our dire need. It is not sufficient to prepare the students to meet competition also prepares them to face and manage academic stressors with emotional stability.

Therefore, the current study aimed to investigate the impact of CBT on adolescents facing academic problems. This study mainly focuses on procrastination, academic burnout, text anxiety, and self-handicapping, which increases emotional disturbance and decreases students' academic performance. In addition, it significantly affects students functioning and performance at school. Furthermore, this study also explores the impact of CBT on managing the unrealistic parental expectations that associates with emotional disturbance. The overall impact of this study is to improve students' emotional stability, school functioning and academic achievements among adolescents.

## Research design and methods

### Study design

In this prospective randomized control trial (RCT), we assessed the effectiveness of CBT among adolescents using pre- and post-assessment measures. Participants were taken from different public and private higher secondary schools in the district of Faisalabad, Pakistan, between May 2021 and December 2022. We assessed the outcomes at the baselines and post-interventions. The study protocol was approved by the Institutional Review Board (IRB), Government College University, and Faisalabad, Pakistan (i.e., Ref.No.GCUF/ERC/2268). Furthermore, the protocol was registered and approved by the Thai Clinical Trial Registry (i.e., TCTR20210704003 on 04 July 2021) with the following URL: https://www.thaiclinicaltrials.org/show/TCTR20210704003).

### Participants

In this RCT, $N = 200$ adolescents facing academic problems were enrolled in different higher secondary schools. The study sample size was calculated through G-Power software using the effect size (f) = 0.20, $\alpha = 0.05$, power (1-$\beta$ error prob.) = 0.95 with actual power = 0.96, which structured the sample size of 60 participants for both treatment conditions (i.e., experimental and waitlist control conditions combined). We enrolled $N = 200$ respondents in this study. After eligibility assessment and screening, 129 participants met the eligibility criteria, and 71 were excluded for various reasons that support the exclusion criteria. Eligible participants were equally allocated to experimental (EXP = 64) and waitlist-control conditions (WLC = 65) through random assignment (see Fig. 1; Table 1). Participants' age range was 14 to 18 years. The sample attrition rate was higher, which could be because of exam days.

### Inclusion and exclusion criteria

In this RCT, students who have been experiencing emotional disturbance for at least one week in the past 6 months but whose level of disturbance does not meet the diagnostic criteria of any psychiatric disorder were included in the study. Participants who had a history of psychiatric illness, a history of in-patient treatment, a history of psychotropic medication or a history of counseling/psychotherapy sessions in the past were excluded from the study. In order to cross-check, participants who have high scores (> average score) on the academic burnout, general procrastination scale and anxiety inventory and low scores (< average score) on academic performance were included in the study. Participants were regular higher secondary school students from the pre-medical/pre-engineering group having no history of dropout and academic failure in past examinations. Participants whose parents have severe medical/psychiatric problems and who are under treatment were also excluded from the study. Those participants who have single parents/were living with single parents and parents who have separated/divorced were also excluded. Participants with any disability were also excluded from the study.

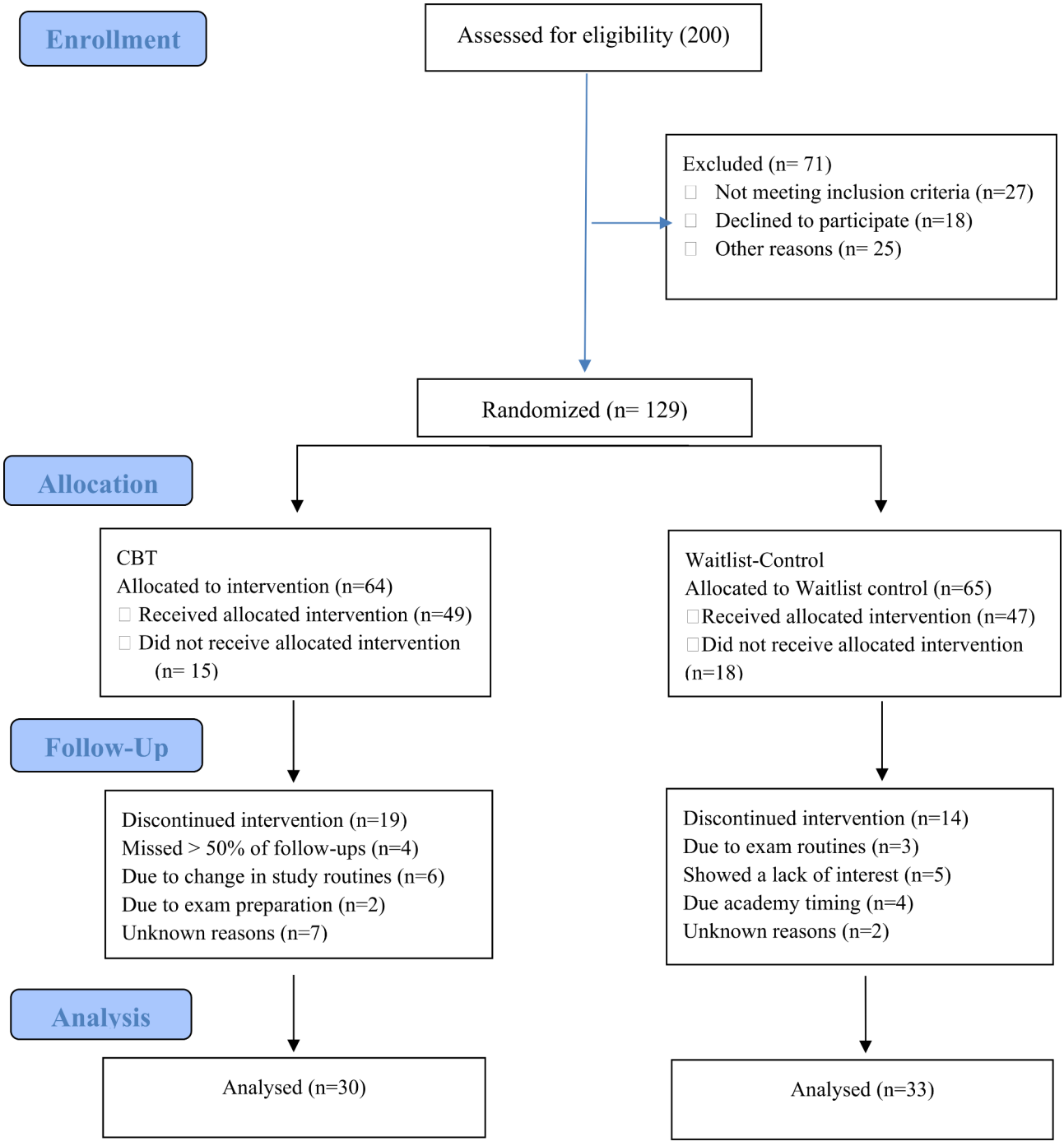



**Fig. 1** Flow diagram of students who have high to moderate parental expectations ($N = 200$)

#### Eligibility assessment and screening

Initially, the researcher contacted the class teachers and received data on adolescents facing academic difficulties. A structured clinical interview was conducted with the participants to investigate the nature and severity of the problem. After getting in-depth information, a brief interview was conducted with the parents and class teachers to cross-check the information. Participants who were facing academic difficulties and causing emotional disturbance were considered for further eligibility assessment and screening. Moreover, a clinical psychologist administered baseline measures, and participants who scored high (> average score) were selected for the study.

**Table 1** Comparison of participants' demographic characteristics between experimental waitlist control condition (*N* = 129)

| Groups | | Overall | Experimental | Waitlist Control | $X^2$/t | *P* |
|---|---|---|---|---|---|---|
| | | | M(SD) | M(SD) | | |
| N Allocated | | 129 | 64(49.6%) | 65(50.3%) | | |
| N Drop | | 78 | 37(47.4%) | 41(52.6%) | | |
| N Final | | 63 | 30(47.6%) | 33(52.3%) | | |
| Age M(SD) | | 17.58(1.86) | 17.69(1.91) | 17.46(1.80) | | |
| Siblings | | 3.87(1.61) | 4.07(1.65) | 3.68(1.55) | | |
| Monthly Income | | 87403.10(129394.77) | 92698.41(142060.94) | 82348.48(116909.62) | | |
| Gender | Male | 63(48.8%) | 31(49.2%) | 32(48.5%) | 0.065 | 0.799 |
| | Female | 66(51.2%) | 32(50.8%) | 34(51.5%) | | |
| Father Education | Metric | 42(32.6%) | 20(31.7%) | 22(33.3%) | 0.140 | 0.933 |
| | FA | 43(33.3%) | 21(33.3%) | 22(33.3%) | | |
| | Graduation | 44(34.1%) | 22(34.9%) | 22(33.3%) | | |
| Father Occupation | Govt. | 68(47.3%) | 31(49.2%) | 37(56.1%) | 0.608 | 0.436 |
| | Business | 61(52.7%) | 32(50.8%) | 29(43.9%) | | |
| Mother Education | < Metric | 42(32.6%) | 20(31.7%) | 22(33.3%) | 9.71 | 0.008 |
| | Metric | 47(36.4%) | 16(25.4%) | 31(47.0%) | | |
| | > Metric | 40(31.0%) | 27(42.9%) | 13(19.7%) | | |
| Mother Occupation | Non-working | 65(50.4%) | 31(49.2%) | 34(51.5%) | 0.069 | 0.793 |
| | Working | 64(49.6%) | 32(50.8%) | 32(48.5%) | | |
| Family system | Nuclear | 70(54.3%) | 38(60.3%) | 32(48.5%) | 1.38 | 0.239 |
| | Joint | 59(45.7%) | 25(39.7%) | 34(51.5%) | | |
| Socioeconomic | Lower | 11(8.5%) | 9(14.3%) | 2(3.0%) | 5.62 | 0.060 |
| | Middle | 112(86.8%) | 52(82.5%) | 60(90.9%) | | |
| | High | 6(4.7%) | 2(3.2%) | 4(6.1%) | | |
| Residence | Urban | 67(51.9%) | 37(58.7%) | 30(45.5%) | 2.27 | 0.131 |
| | Rural | 62(48.1%) | 26(41.3%) | 36(54.5%) | | |
| Institute | Govt. | 69(53.5%) | 32(50.8%) | 37(56.1%) | 0.359 | 0.549 |
| | Private | 60(46.5%) | 31(49.2%) | 29(43.9%) | | |
| Education | 9 Years | 29(22.5%0 | 18(28.6%) | 11(16.7%) | 3.09 | 0.377 |
| | 10 Years | 30(23.3%) | 12(19.0%) | 18(27.3%) | | |
| | 11 Years | 37(28.7%) | 17(27.0%) | 20(30.3%) | | |
| | 12 Years | 33(25.6%) | 16(25.4%) | 17(25.8%) | | |
| Participation in | Minor level | 34(26.4%) | 14(22.2%) | 20(30.3%) | 3.37 | 0.183 |
| Extracurricular | Average | 51(39.5%) | 30(47.6%) | 21(31.8%) | | |
| Activities | Above average | 44(34.1%) | 19(30.2%) | 25(37.9%) | | |

### Feasibility

Initially, the study protocol and its significance were discussed with the school administration. It was briefed to the institutions that our purpose of this study is to design a treatment protocol to address psychological disturbance among students which is increasing gradually among our academic institutions. The teaching staff appreciated and agreed with our study protocol, and a formal study procedure was started. Informed consent was taken from parents and then students were referred for psychological evaluation. Moreover, it was assured that the student's identities would not be disclosed, and all the received information would remain confidential.

### Setting

The study was carried out at five different schools in Faisalabad, Punjab, Pakistan. The students received counselling sessions each week during their tutorial period (a period that is customarily fixed in each week's timetable in Pakistani schools for any co-curricular activities) in the designated room for the school counsellor. Six to eight sessions were provided to each student in the treatment conditions. Masters' students of clinical psychology (with expertise in CBT) provided these counseling sessions (under the supervision of their respective teachers).

### Randomization

We randomly assigned (1:1) the participants across the treatment conditions through random assignment using block randomization. Furthermore, the chi-square test was used to formulate matchable groups; a *p*-value close to 1 considered the group highly matchable. Similarly, the independent sample t-test matched the groups on the

continuous variables. All the participants were blinded to identify their assigned group for the treatment conditions. Researchers informed the participants that they are being randomized to receive treatment, and the goal is to reduce students' level of academic burnout and procrastination that is causing an emotional disturbance.

### Procedure

For this RCT, the participant enrollment process was in July 2021, and the trial was completed in December 2021, at different higher secondary schools in Faisalabad. After getting permission from the schools' authorities and signed informed consent from their patients, participants were registered for this clinical trial. Those respondents who met the study eligibility criteria were called for further baseline assessment. Participants' baseline assessment was completed in the school setting in one-on-one sitting. Eligible participants were allocated to treatment conditions using random assignment through block randomization. Participants were assessed at baseline assessment (pre-assessment) and after the treatment (post-assessment), with a duration of 3 months between baseline and treatment completion. We conducted 6–8 CBT-based therapeutic sessions. Each session was structured based on a special agenda and goal setting (see interventions). The same baseline and post-assessment procedures were followed for both treatment conditions. Sessions were planned one session per week in one-on-one sitting during the tutorial period in the office of the student's counsellor, maintaining privacy and confidentiality.

### Interventions

Cognitive behavior therapy is an effective treatment intervention for adolescents facing academic problems, and it has produced sufficient treatment outcomes. It is considered an effective and reliable therapy to address emotional, behavioral and cognitive problems. After investigating some relevant evidence, a structured treatment protocol was designed considering the value of the cultural context. This CBT-based treatment protocol was tailored from the book of Cognitive Behaviour therapy-basic and Beyond latest edition [60]. The structured treatment protocol is based on 6–8 sessions with a duration of 60 min and once a week. The designed treatment protocol was tested on 3–4 participants and findings reported that the designed protocol is fulfilling our need. The treatment plan was comprised of the following main agendas, i.e., *psychoeducation*, *conceptualization and cognitive restructuring*, *stress management*, *behavioral/emotional management*, *problem-solving skills*, *effective parent-child relationships*, *goal-ordinated training and laps-relapse prevention*. This process started with a briefly description with the participants about advantages of the protocol and then therapist moved to the psychoeducation to the participants about the problems. Furthermore, the therapist explained the significance that you will be able to cope with emotional disturbance and other outcomes; such as, procrastination [56], burnout [54, 55], anxiety and self-handicapping behavior which reduce academic performance [58]. Moreover, all sessions were designed to address the target problems using proper techniques and homework assignments were given to the participants to practice and strengthen the modified thoughts, behaviours, emotions and briefs. The main treatment goals were psychoeducation, conceptualization and cognitive restructuring, stress management, behavioral/emotional management, problem-solving skills, effective parent-child relationship, tasks management and lapse and relapse prevention (see Supplementary File- Table A).

**Waitlist control condition** Participants allocated to the condition received no treatment for 8 weeks. Their pre- and post-assessment was completed within the same time interval as the experimental group.

### Assessment measures

**Demographic form** The demographic form was used to attain personal information from students, such as age, education, socioeconomic status, family system, total family members, marital status, class, and institutional affiliation. In addition, the information related to their parents, such as the father's education, father's occupation, mother's education and mother's occupation, was also asked in the demographic form. This form was used for further analysis and to tabulate results.

#### *Primary outcome measures*

*Maslach burnout inventory-student survey (MBI-SS)* MBI-SS was developed to assess the degree of burnout among students [61]. The MBI-SS contains 15 items that investigate three dimensions, such as emotional exhaustion (5 items), cynicism (4 items) and academic efficacy (6 items). It is scored on a 7-point Likert scale from "Never = 1" to "Always = 7." The researchers report adequate internal consistency of the measure with a coefficient alpha of 0.80 for emotional exhaustion, 0.86 for cynicism, and 0.67 for academic efficacy.

*The General Procrastination Scale (GPS) investigates the procrastination level* among students [62]. The GPS is comprised of 20 items with no subscales. The GPS is a 5-point Likert scale scored from *"Strongly agree = 5"* to *"Strongly disagree = 1"*. Cronbach's alpha coefficient is estimated at 0.77, which indicates good internal consistency of the scale. The scale measures the global tendency to procrastinate across various daily tasks.

*Test Anxiety Inventory (TAI)*: The TAI is constructed to assess the degree of anxiety related to tests/exams in students [63]. The TAI is comprised of 20 items with three sub-domains: Test Anxiety-Total (TA-T), Test Anxiety-Worry (TA-W), and Test Anxiety-Emotionality (TA-E). The TAI is a 4-point Likert-type scale, and the students have to respond to the four options: (1) Almost Never, (2) Sometimes, (3) Often and (4) Almost Always. The TAI has internal consistency for the whole as well as for the subscale: 0.96 for TA-T, 0.91 for TA-W and 0.91 for TA-E.

*The Academic Self-Handicapping Scale (ASHS);* The ASHS is comprised of 25 items to assess self-handicapping behavior among students [64] originally developed it. The ASHS assesses the tendency of lack of effort, simulation, laziness, emotional uneasiness, and worry about progress which associates handicapping behaviour. The ASHS is scored on a 5-point Likert scale ranging from *"Very little = 1"* to *"Very much = 5"*. Its internal consistency is estimated at 0.80, which reflects it is highly reliable. The internal consistency for making excuses is estimated at 0.60 and for bad temper is 0.72.

*Students' Academic Performance (SAP)*: The purpose of the SAP was to estimate the students' previous and current academic performance. We assessed the student's academic performance on the base of grades three times. For example, (a) students' academic performance in the past year's final exam, (b) three months' students' academic performance made by class teachers on the basis of monthly tests and (c) students' academic performance was estimated on the basis of final exam conducted 3–4 months after treatment. We converted the student's grades into percentages (%) between 1 and 100% using a formula made by the Board of Intermediate and Secondary Education department (i.e. Grade A + = 80% or above, A = 70–79%, B = 60–69%, C = 50–59% and D = 40–48% marks). This grade system information is available at https://bisegrw.edu.pk/grading.html.

*Students' attendance record (SAR)* The purpose of the SAR was to estimate the students' change in attendance records in the previous and current academic year to estimate students' functioning at school. We calculated 4-times students' attendance records in order to estimate the regularity of the students. Initially, we collected attendance records before 1 year (12 months before baseline), current 3 months (3 months completed at time of baseline), students' attendance records during the treatment and 3 months attendance records after completion of treatment. We calculated the students' attendance on the basis of percentage (%) between 1 and 100%.

### Secondary measure

*Living up to Parental Expectation Inventory (LPEI)*: The LPEI is developed to assess parental unrealistic expectations among students [65]. It is comprised of 25 items with two subscales, Personal Maturity and Academic Achievement. Each statement scores on a 6-point Likert scale. High scores indicate high perceived parental expectations and low scores indicate low perceived parental expectations; the scale has yielded a good reliability of 0.87.

### Ethical considerations

In this RCT, all the ethical considerations recommended by the review board were followed. Initially, informed consent was taken from the parents, who briefly described the study's significance. The therapist assured the participants that the received information would remain confidential and that participants' identities would never be disclosed. Participants were also informed that if they felt discomfort or unease at any research stage, they had the right to withdraw from the study.

### Statistical analysis

After enrollment, participants who met the inclusion criteria were allocated to treatment conditions. The Chi-square test was performed to ensure the matchable group characteristics on the categorical variables, and an independent sample t-test was used for the continuous variables. Chi-square and t-test yield insignificant differences ($p$-value > 0.05) between treatment groups on categorical and continuous variables. Repeated measure ANOVA statistic was used to investigate the group difference, time interval and group x time on the pre-and post-assessment conditions. The parameters set for the analysis were alpha 0.05, confidence interval 0.95%, effect size 0.20, and $p$-values < 0.05 for all analyses. All the statistical findings were obtained using IBM SPSS-Version. 26.0.

## Results

### Recruitment and attrition

in this RCT, a sample of 200 participants was recruited, and 129 met the eligibility, they were allocated to experimental 64(50%) and waitlist control condition 65(50%). Attrition rate was 78(60.5%) and retention rate was 63(48.8%). All the demographic characteristics were calculated using chai-square and t-test, and found that both groups were matchable and insignificant differences were calculated (see Table 1).

### Primary outcomes

We found a significant mean difference between EXP and WLC conditions in post-testing GPS scores, indicating that 0.57% CBT substantially decreased procrastination among students. Furthermore, significant mean differences were found between baseline and post-testing scores on the BES, BCS, and AES scale between EXP and

**Table 2** Mixed-design ANOVA statistics comparing Procrastination, Burnout, Self-handicapping behavior and anxiety between Pre-post Assessment of Experimental and Control group among adolescents

| Variables | Experimental group | | Control group | | | | |
|---|---|---|---|---|---|---|---|
| | M(SD) | M(SD) | M(SD) | M(SD) | F | *P* | ηp2 |
| GPS | 59.23(11.09) | 34.93(5.65) | 56.06(5.08) | 56.73(4.79) | 53.16 | 0.001 | 0.573 |
| BES | 15.7(5.28) | 3.33(1.58) | 14.8(5.33) | 17.0(10.55) | 11.08 | 0.001 | 0.36 |
| BCS | 13.5(4.57) | 5.27(2.75) | 12.62(4.33) | 12.31(4.58) | 25.71 | 0.001 | 0.31 |
| AES | 19.93(4.90) | 27.33(4.40) | 18.84(6.03) | 18.21(5.76) | 26.01 | 0.001 | 0.39 |
| ASHS | 81.8(11.09) | 68.23(11.48) | 82.17(10.57) | 82.60(9.51) | 10.17 | 0.001 | 0.16 |
| TAI-ES | 27.6(2.54) | 12.93(2.92) | 21.21(4.77) | 18.30(5.72) | 77.64 | 0.001 | 0.56 |
| TAI-WS | 26.70(3.06) | 11.63(2.07) | 21.67(4.25) | 18.81(4.77) | 79.37 | 0.001 | 0.56 |
| TAI-TS | 14.33(1.24) | 7.87(0.94) | 10.42(2.01) | 9.18(2.54) | 75.86 | 0.001 | 0.55 |
| TAI | 48.63(5.87) | 32.43(4.45) | 47.30(9.52) | 46.30(11.98) | 88.17 | 0.001 | 0.59 |

*Note: ηp2 = Partial Squared; GPS: General Procrastination Scale; BES: Burnout Exhaustion Subscale; BCS: Burnout Cynicism Subscale; BAE: Academic Efficacy Subscale; ASHS: Academic Self-Handicapping Scale; TAI-ES: Test Anxiety Inventory-Emotional Subscale; TAI-WS: Test Anxiety Inventory-Worry Subscale; TAI-TS: Test Anxiety Inventory-Total Subscale; TAI: Test Anxiety Inventory*

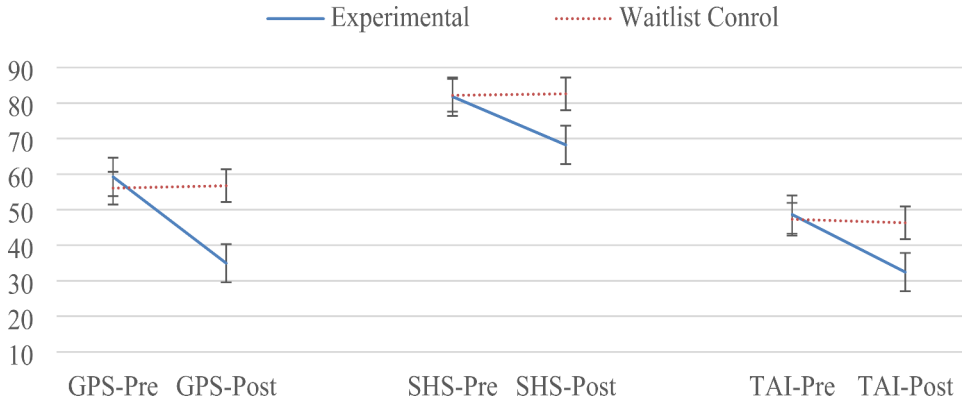



**Fig. 2** Impact of CBT on GPS, SHS and TAI over the course of treatment between EXP and WLC conditions. *Note: CBT = Cognitive Behavior Therapy; GPS = General Procrastination Scale; SHS = Self-Handicapping Scale; TAI = Test Anxiety Inventory*

WLC, indicating CBT reduced 36% burnout exhaustion, 31% burnout cynicism, and 39% burnout. CBT improved the level of academic efficacy. Similarly, the EXP group was significantly different compared to WLC on overall ASHS, which shows that CBT 16% improved self-handicapping behavior among adolescents. Analysis reveals a significant difference between EXP and WLC after CBT on TAI-ES, TAI-WS, TAI-TS, and overall TAI, which shows CBT sessions decreased the level of test anxiety emotional of 56%, decreased worry by 56%, anxiety by 55%, and overall test anxiety to the degree of 59% (see Table 2).

The findings reported that participants who received the allocated treatment significantly decreased GPS, SHS and TAI compared to those who never received the allocated treatment conditions. This reflects that CBT is an evidence-based treatment intervention that has the potential to address students' emotional problems (see Fig. 2).

The findings indicate that CBT significantly decreased the level of BES and BCS in the experimental group compared to the waitlist control. In addition, CBT significantly enhanced the chance of AES among students who received the treatment as compared to those who did not receive the allocated interventions (see Fig. 3).

It shows that treatment intervention significantly helped the adolescents overcome their emotional problems, kept them motivated, and encouraged them to focus on their target goals. This reflects that those students who received the treatment significantly improved their academic performance as compared to those who did not receive the treatment (See Fig. 4).

The study finding shows that cognitive behaviour therapy substantially improved the level of students' academic performance among adolescents. The SAP in the previous final exam and in the past 3 months was found not significantly different because students' emotional problems started and gradual decline was observed in the academic performance. Similarly, insignificant difference was found between experimental and control group while after treatment significant difference was found between the experimental and control group and significant improvement was seen in the academic performance among experimental group after follow up (see Table 3).

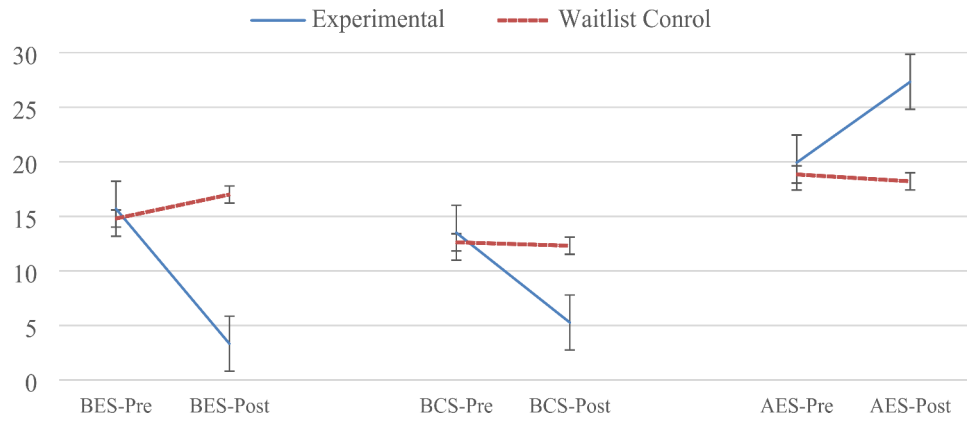



**Fig. 3** Impact of CBT on BES, BCS and AES throughout treatment between EXP and WLC conditions. *Note: CBT = Cognitive Behavior Therapy; BES = Burnout Exhaustion Subscale; BCS = Burnout Cynicism Subscale; AES = Academic Efficacy Subscale*

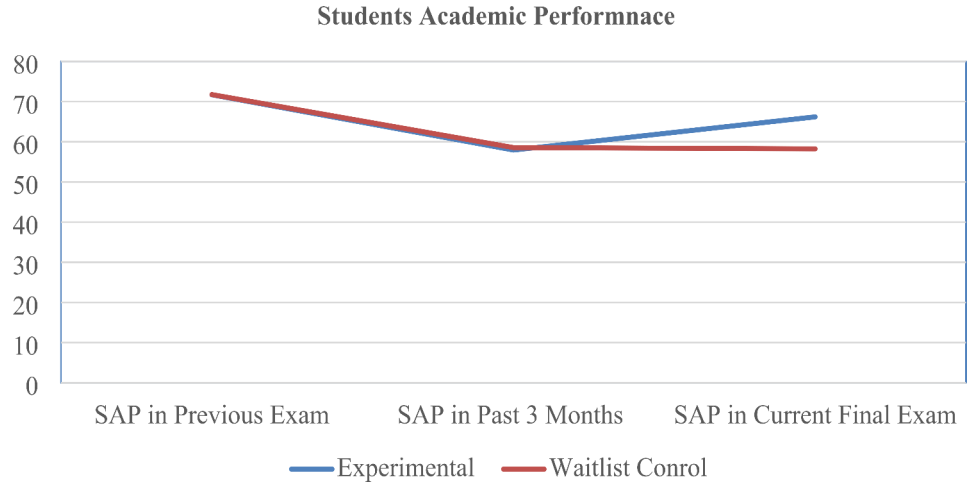



**Fig. 4** Impact of CBT on students academic performance over the course of treatment between EXP and WLC conditions

**Table 3** Mixed-design ANOVA statistics for students' academic performance (SAP) in the previous final exam, at baselines (3 months performance) and current final exam (after 3 months follow up) among adolescents

| Groups | Experimental Group | | | Waitlist Control | | | F | *P*-Value | $\eta_p^2$ |
|---|---|---|---|---|---|---|---|---|---|
| | SAP in Previous Final Exam | SAP in Past 3 Months | SAP in Current Final Exam | SAP in Previous Final Exam | SAP in Past 3-Months | SAP in Current Final Exam | | | |
| | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | | | |
| Students' Academic Performance | 71.07(8.95) | 57.94(6.15) | 66.20(4.38) | 71.75(5.95) | 58.55(5.36) | 58.24(5.32) | 280.12 | 0.001 | 0.821 |

Finding reveals that cognitive behaviour therapy significantly improved the school function among adolescents. For example, one year before students' average attendance was 78.43% (experimental group) and 76.79% (control group) which indicates it was good attendance record and no significant difference exist. Similar, case is with 3 months before at baseline but attendance decline to 57.93% and 57.90%. This indicates students' emotional disturbance reduced the school functioning. As treatment was started the students attendance record was increased to 66.40% and it significantly improved at follow up almost 69.27% but control group experienced the same problems and their attendance remained the same.

The results showed that after CBT students attendance rate significantly improved which is almost close to their one year back attendance rate while those who did not received the treatment their attendance remained declined due to their emotional and behavioral problems (see Fig. 5).

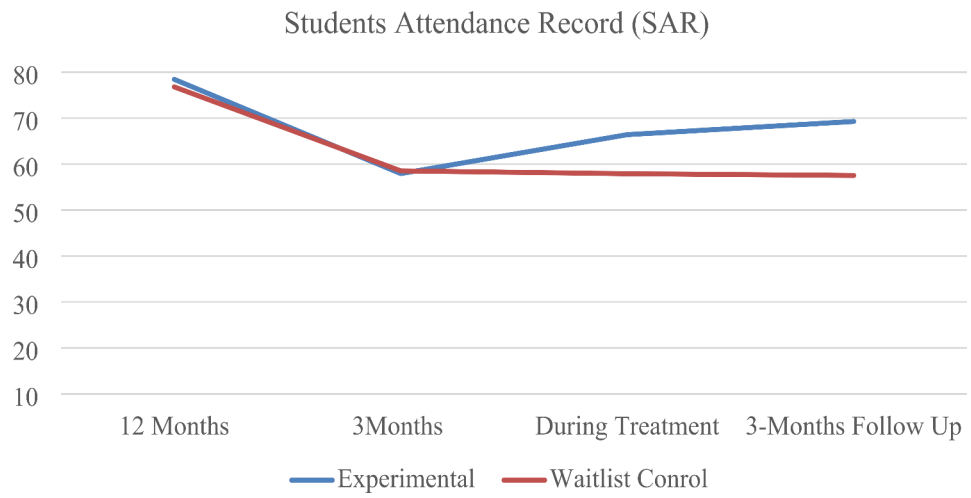



**Fig. 5** Impact of CBT on students attendance record over the course of treatment between EXP and WLC conditions

**Table 4** Mixed-design ANOVA statistics for students' attendance record (SAR) of past 12 months, 3 months before baseline, during treatment and 3 months after treatment among adolescents

| Groups | Experimental Group | | | | Waitlist Control | | | | F | *P*-Value | $\eta_p^2$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | SAR 12 Months Before | SAR 3 months Before Baseline | SAR During Treatment | SAR 3 Month After Treatment | SAR in 12 Months Before | SAR 3 months Before Baseline | SAR During Treatment | SAR 3 Month After Treatment | | | |
| | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | M(SD) | | | |
| Students Attendance Record | 78.43(9.86) | 57.93(6.15) | 66.40(4.86) | 69.27(6.74) | 76.97(10.50) | 58.52(4.62) | 57.90(4.95) | 57.52(5.15) | 239.32 | 0.001 | 0.797 |

**Table 5** Mixed-design ANOVA statistics comparing living up to parental expectation between pre-post assessment of experimental and control groups among adolescents

| Variables | Experimental Group | | Waitlist Control group | | | | |
|---|---|---|---|---|---|---|---|
| | M(SD) | M(SD) | M(SD) | M(SD) | F | *P* | ηp2 |
| PMP | 5.05(1.17) | 4.60(0.31) | 5.34(0.43) | 4.59(0.32) | 29.85 | 0.001 | 0.33 |
| PMS | 4.38(0.32) | 3.18(0.99) | 4.25(0.29) | 2.83(0.80) | 127.90 | 0.001 | 0.68 |
| AAP | 4.93(1.20) | 4.82(0.37) | 5.47(0.47) | 4.96(0.28) | 7.68 | 0.001 | 0.11 |
| AAS | 4.00(0.40) | 3.17(1.10) | 4.20(0.30) | 2.94(0.82) | 64.02 | 0.001 | 0.51 |
| LEP | 59.87(19.30) | 10.97(5.32) | 62.75(12.21) | 12.27(5.50) | 546.46 | 0.001 | 0.89 |

*Note: ηp2 = Partial Squared; PMP: Personal Maturity Parent; PMS: Personal Maturity Student; AAP: Academic Achievement Parent; AAS: Academic Achievement Student; PPE: Perceived parental Expectation; PSP: Perceived Self-Performance; LPE: Living up to Parental Expectation*

### *Secondary findings*

We found a significant mean difference between EXP and WLC conditions between the baselines and post-testing scores on the perceived parental expectation after CBT among students. For example, CBT 33% increased the chance of PMP, which was lower compared to 68% PMS. Similarly, CBT 51% increased the chance of AAS. In comparison, CBT remained helpful in enhancing the chance of perceived AAP by only 11%. Overall, CBT helped the students manage perceived parental expectations improved by up to 89% among students who received the interventions (see Table 4).

## Discussion

Our findings reported that CBT substantially managed procrastination, burnout, test anxiety and self-handicapping behaviour among adolescents [66, 67]. These findings are consistent with the findings of the previous studies [68]. Persistent academic burnout and procrastination behaviour were seen among the adolescents, and they were suspected of clinical symptoms if they were not treated [69]. CBT effectively addressed the depressive symptoms after academic problems in good manners, and adolescents actively participated in this experiment ( [70]. CBT also remained helpful in maintaining positive attitudes among adolescents and motivation toward

treatment and academic performance. Symptoms in adolescents significantly reduce academic performance [71]. CBT developed an insight after psycho-educating them about the problems, and they were guided and trained to understand, regulate and manage their emotions through modifying beliefs [72]. Post-assessment findings ensure that participants were observed to have learned the decision-making skills and priority setting to manage their daily living tasks with positive thoughts, emotions and beliefs [73].

The study finding indicates that CBT reduced the level of procrastination among adolescents, helped them manage procrastination and increased the chance of academic performance in the current circumstances [56]. Procrastination leads to test anxiety, and self-handicapping behaviour influences academic performance [74]. Students usually avoid daily homework assignments and academic tasks seeking relaxation from the current routines, but with time, behaviors turn into poor motivations and negative attitudes [75]. Students use procrastination in different forms, such as lack of interest, lack of priority setting and active decision-making skills, unnecessary delay in completing academic assignments, poor time management, excessive fear of failure, high expectations for grades, poor self-regulation and peer pressure [76]. CBT effectively helped the students manage daily tasks and sustain motivation and positive attitudes toward academic activities. Moreover, CBT sessions provide awareness about the root cause of problems. They developed an insight into how these issues can be resolved easily and productively, ultimately decreasing the degree of procrastination.

Moreover, our study findings reported that a significant mean difference on the self-handicapping scale was found in the experimental group than in waitlist control; this reflects that CBT reduced self-handicapping among adolescents [58]. Self-handicapping leads to procrastination and affects academic achievement, but CBT-based sessions successfully managed self-handicapping behavior and increased academic achievement [59]. Self-handicapping activities and illogical beliefs negatively impact goal-setting behavior and general functioning [77]. These findings align with what has been found in earlier investigations [59]. Irrational ideas and self-handicapping acts are both significant phenomena for general mental functioning. Irrational perceptions regarding approval were found to impact self-handicapping in the study substantially. The ABC model explains to the students that strong coping easily controls self-handicapping and other emotional behaviours [78]. Self-handicapping leads to procrastination, then students' academic achievement decreases; after the treatment, self-handicapping behaviour reduces and increases academic achievement [59].

The study findings indicate that test anxiety levels were high at baseline screening. A significant decrease in test anxiety scores was found in post-assessment, showing that CBT effectively treated test anxiety among adolescents [79]. In addition, CBT sessions helped the participants to understand and manage their irrational fears about the outcomes of the exam. Also, they gave an insight into regulating them with a positive attitude using planning and strategies [80]. Test anxiety causes fears, gradually developing an irrational belief among students that they are incompetent. They never succeed in academic outcomes, but CBT altered these beliefs and cognition into goal-directed channels, and CBT sessions remain productive in managing emotions, worries, and overall academic anxieties [58]. Before final exams, students usually perceive academic anxiety, which provokes fear of failure and difficulty in their studies [74]. In the long run, it changes into burnout factors (cynicism, emotional exhaustion and academic efficacy), which correlates with distress [81], psychological difficulties and poor academic performance [69].

Furthermore, the analysis reveals that high parental expectations perceived by students were manageable in the experimental group compared to a waitlist control group after practicing CBT-based sessions with the adolescents [82]. High parental expectation correlates with distress in students when they think their academic achievements would not be considered appreciated if they did not meet the parental expectations. On the other hand, parents have realistic expectations correlates, which do not correlate with frustration and emotional disturbance [83]. Further analysis reveals that parents are important in improving and sustaining students' performance. Various positive parenting variables are seemingly positive in stable personality development, and negative factors correlate with emotional problems. Sometimes irrational parental expectations correlate with self-handicapping behaviour and procrastination [84]. Parental psychological control impacts adolescents' behavioural problems and academic achievements [83].

## Conclusion

In this study, CBT was conducted on adolescents experiencing symptoms of burnout, procrastination and self-handicapping behaviour. It is concluded that CBT's significant impact on adolescents was investigated by addressing the symptoms related to procrastination, burnout, and self-handicapping behaviour. In addition, it was seen that anxieties and fears related to tests and exams were noted lower in the experimental group as compared to the control group after CBT. CBT sessions also helped the students understand and manage parental expectations realistically and logically. Overall, students'

academic achievement was also observed to be higher after the trial in the experimental group.

### Limitations of the study

In Pakistan, the tendency of emotional disturbance among adolescents is rising due to high academic competition and lack of financial resources. In light of these issues, the present study is planned as a new initiative with students in public and private higher secondary schools. However, teachers and students showed greater interest in studying but their interest gradually reduced due to study workload. secondly, this counseling program was not compulsory to adopt. This was the main reason for termination. Second, the main reason was the lack of awareness among teachers and students about mental health problems in a school setting. Therefore, this was an addition to the school setting, and some students avoid to take part/leaving the program because they felt that they would be labialized. Their friends will think negatively about them while everything is confidential. Another limitation of this program was that it was implemented with science students only. Moreover, we only target the students who are facing emotional disturbance, so this counseling program does not describe the students having serious mental health issues and their solutions. This program was not compulsory for students, but it was arranged at students' convenience. Therefore, some difficulties were faced with running it smoothly. For example, if this counseling program were compulsory and planned over a specific time with a particular duration and interval, it would be less burdened, more productive, comprehensive, and easy to implement. This counseling program was not tested on students from rural areas where they have more lack of awareness, and emotional problems were another limitation of this program. Some parents and schools do not allow us because they understand our program will take time and can affect students' education; if they allow us, we will be able to manage students' emotional problems as well as implement this counseling program on a larger scale.

### Recommendations and future directions

CBT-based counselling programs produced very productive and effective outcomes. On the basis of this protocol, it is recommended that there would be a longitudinal study that will assess the outcomes over time and will help to manage students' emotional problems for students' positive personality growth and development. Considering the outcomes, the state should plan a comprehensive program and higher school psychologists for schools to spread awareness and train the teachers, parents and students about mental health issues would be the complete and comprehensive solution. This counselling program provides a baseline, and it will help to tailor a culturally adapted comprehensive counselling program for our school children, which will be more effective and productive. This should be categorised into different levels, and it should start with preschool children, and it will continue till higher secondary school education; it will bring a remarkable change in students' personality growth, mental stability, psychological well-being and high academic achievements.

### The implication of the study

**Students** Students usually perceive stress when they struggle hard for high achievements and grades. Sometimes they feel stressed due to peer group competition and sometimes because of irrational and unrealistic parental expectations. This study suggests that when students feel they are under stress and their performance is decreasing or expect any stressful situation to worsen in the coming days, they must consult a counselor and seek help and guidance to overcome the problematic situations at an early stage.

**Parents** The present study endorses that parents should have realistic expectations about children's grades and support them if they observe any weakness in the child. Unrealistic and irrational parental expectations could correlate with burnout, procrastination and handicapping behaviors in adolescents, significantly influencing children's mental health and academic performance.

**Teachers** This study also recommends that teachers should give required educational exposure after understanding students' capacity and intellectual level with manageable and doable tasks and assignments. Teachers should support and cooperate with students as it gives confidence, encouragement and motivation to students, ultimately promoting their academic performance.

**Institutions** It is suggested that institutions should develop proper counseling centers in the academic institutions that must counsel students related mental health, vocational, and educational problems. Dysregulated students are usually more vulnerable to developing mental health problems. Furthermore, due to a significant decrease in economic growth and increased unemployment and poverty in Pakistan, protecting the students at their educational institutes has become paramount. Care and support can easily protect and keep them consistent about their goals.

### Abbreviations

| | |
|---|---|
| RCT | Randomized Control Trial |
| CBT | Cognitive Behavioral Therapy |
| GPS | General Procrastination Scale |
| BES | Burnout Exhaustion Subscale |
| BCS | Burnout Cynicism Subscale |
| BAE | Academic Efficacy Subscale |

| | |
|---|---|
| ASHS | Academic Self-Handicapping Scale |
| TAI-ES | Test Anxiety Inventory-Emotional Subscale |
| TAI-WS | Test Anxiety Inventory-Worry Subscale |
| TAI-TS | Test Anxiety Inventory-Total Subscale |
| TAI | Test Anxiety Inventory |
| PMP | Personal Maturity Parent |
| PMS | Personal Maturity Student |
| AAP | Academic Achievement Parent |
| AAS | Academic Achievement Student |
| PPE | Perceived parental Expectation |
| PSP | Perceived Self-Performance |
| LPE | Living up to Parental Expectation |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40359-025-02371-2.

Supplementary Material 1

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**Author contributions**
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**Data availability**
The dataset generated and/or analyzed during the present study is not publicly available because no permission was taken from the participants and the college administration where the study was conducted. The datasets are available from the corresponding authors on a special request.

## Declarations

**Ethics approval and informed consent**
The study protocol was approved by the Institutional Review Board (IRB), Government College University, and Faisalabad, Pakistan (i.e., Ref.No. GCUF/ERC/2270). Furthermore, the protocol was also registered and approved by the Thai Clinical Trial Registry (i.e., TCTR20210703002 on 03 July 2021, with URL: https://www.thaiclinicaltrials.org/show/TCTR20210703002). Moreover, informed consent was obtained from all the participants. Participants who read and signed the consent form were enrolled in the study. All procedures, including participant enrollment, screening, data collection, data scrutiny and analysis, were performed in accordance with relevant guidelines.

**Consent for publication**
Not applicable.

**Competing interests**
The authors declare no competing interests.

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**Disclosure**
None.

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